

Problem 2.9

```
clear, clc
%{
Frank Sucansky, AERO 1610, Lab 2
Write an expression refers to only the elements that have odd-numbered subscripts in
a vector, regardless of the length of the vector
%}
Vec_A = [1,2,3,4,5,6,7,8]; %create a vector with 8 elements
Vec_Odd = Vec_A(1:2:end); %find odd subscripts in Vec A
Vec_Even = Vec_A(2:2:end); %test that method works for even subscripts aswell
disp(Vec_Odd) %display all odd subscripts
```

Output :

```
1 3 5 7
```

Problem 3.5

```
clear, clc
%{
Nathaniel Moberg, AERE 1610, Lab2
Purpose: Experiment, in the Command Window, with using the fprintf function for real numbers.
Make a note of what happens for each. Use fprintf to print the real number 12345.6789.
• without specifying any field width
• in a field width of 10 with 4 decimal places
• in a field width of 10 with 2 decimal places
• in a field width of 6 with 4 decimal places
• in a field width of 2 with 4 decimal places
%}

A = [12345.6789]; %sets the value as a 1x1 matrix --> in this case works like a variable
fprintf('%f', A); %prints the number with no predetermined field width or number of decimal places
fprintf('\n %10.4f', A); %prints with field width of 10 to 4 decimal places
fprintf('\n %10.2f', A); %prints with field width of 10 to 2 decimal places
fprintf('\n %6.4f', A); %prints with field width of 6 to 4 decimal places
fprintf('\n %2.4f', A); %prints with field width of 2 to 4 decimal places
```

Output :

```
12345.678900
12345.6789
 12345.68
12345.6789
12345.6789>>>
```

Problem 2.32

```
clear,clc
% { Purpose: create a 3x5 matrix and find max values
A = randi([0,9],3,5) % 3x5 matrix with random values
a = max(A) % finds the max value from every column
row1 = 1
row2 = 2
row3 = 3
maxValRow1 = max(A(row1, :)) %finds max value in row 1
maxValRow2 = max(A(row2, :)) %finds max value in row 2
maxValRow3 = max(A(row3, :)) %finds max value in row 3
overallMax = max(A(:)) % finds the maximum value in the entire matrix
```

Output:

A =

```
6 6 0 1 9
2 7 2 8 0
6 4 9 5 4
```

a =

```
6 7 9 8 9
```

row1 =

```
1
```

row2 =

```
2
```

row3 =

```
3
```

maxValRow1 =

```
9
```

maxValRow2 =

8

maxValRow3 =

9

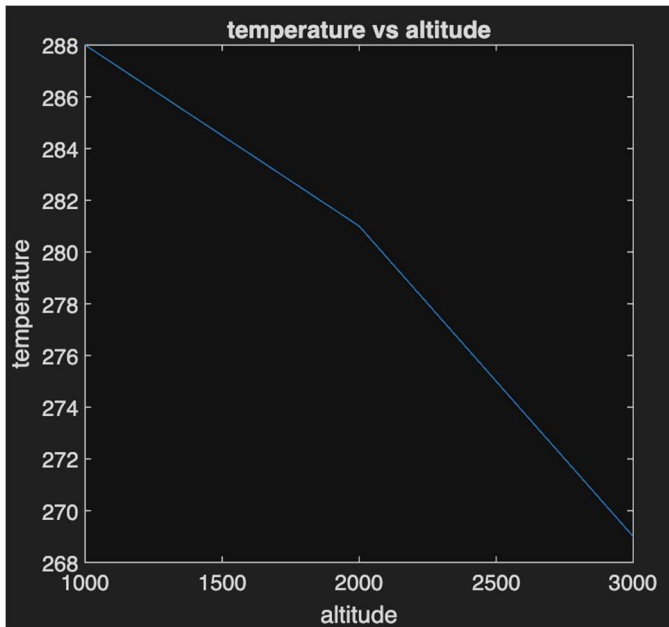
overallMax =

9

Problem 3.17

```
clear, clc
%{
Frank Sucansky, AERO 1610, Lab 2
Write a script that will load alttemps data into a matrix,
separate it into vectors, and then plot the data
with appropriate axis labels and a title.
%}
data = load('alttemps.dat'); % Load the alttemps data
altitudes = data(:,1); % put first column of data into variable alt
temperatures = data(:,2); % put second column of data into variable temp
plot(altitudes,temperatures)
xlabel('altitude')
ylabel('temperature')
title('temperature vs altitude')
```

Output :



Problem 3.7

```
clear,clc
%{
Nathaniel Moberg, AERE 1610, Lab2
Purpose: Write a script that will prompt the user for the number of rows and columns of a matrix,
create a matrix with that many rows and columns of random integers, and write it to a file.
%}
rows = input('Number of Rows: ') %sets number of rows to an input
columns = input('Number of Columns: ') %sets number of columns to an input
randi([1,100],rows,columns) %random integers at a specified range, with the dimensions rowsxcolumns
```

Output :

Number of Rows: 3

rows =

3

Number of Columns: 2

columns =

2

ans =

49 71

45 76

65 28

>>

Problem 6

clear, clc

%{

Frank Sucansky, AERO 1610, Lab 2

1. Load the data from the loadstress_temp.dat file into a matrix.

2. Separate the matrix into vectors representing load and stress at each temperature.

3. Plot stress versus load for both temperatures on the same figure using distinct markers or line styles.

%}

data = load('loadstress_temp.dat');

load = data(:,1);

stress_Temp1 = data(:,2);

stress_Temp2 = data(:,3);

plot(stress_Temp1,load,'r', stress_Temp2,load,'b');

xlabel('stress temperatures')

ylabel('load')

title('load vs stress temperature')

legend('stress temp 1', 'stress temp 2')

Output :

